Pathwise optimal tracking of a fractional Ornstein-Uhlenbeck process in a dynamic portfolio problem

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Introduction

Primary Goal

To build an optimal portfolio of standard Geometric Brownian Motion (GBM) assets that accurately tracks a fractional stochastic process.

Market Imperfections (Synthetic Tracking)

- **Pure Non-Tradability:** Institutional benchmarks (GDP, CPI, climate aggregates) lack exchange-traded vehicles, creating an incomplete market space.
- **Prohibitive Friction:** Fractional media (B_t^H) inherently suffer from extreme microstructure illiquidity and unsustainable rebalancing costs.
- **Capacity & Regulation:** Large-scale institutional flows trigger severe adverse market impact; rigid regulatory caps (Basel IV, Solvency II) penalize direct exposure.
- **Credit Risk & Horizon Mismatch:** Direct physical locking of capital strips tactical flexibility. Replication via liquid underlyings eliminates counterparty risk through central clearing.
- **Rough Volatility Domain:** Empirical volatility (VIX) exhibits anti-persistent sub-diffusion dynamics ($H < 1/2$), creating severe options-market replication barriers.

The Reference Target

Stochastic Basis ($t \geq 0$)

- Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space with a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$.
- Market assets are driven by a $(n-1)$-dimensional BM: $W_t = (W_{1t}, \ldots, W_{(n-1)t})'$.
- The target is driven by an independent two-sided BM, $\tilde{W}_t$ on $\mathbb{R}$, where $\mathcal{F}_0 = \sigma(\tilde{W}_s : s \leq 0)$.

The Underlying Fractional State

- Let ξ_t be a stationary fractional mean-reverting Ornstein–Uhlenbeck process:

$$d\xi_t = \theta(\mu - \xi_t)dt + \sigma dB_t^H$$

  where B_t^H is a fractional Brownian motion ($H \in (0,1)$) governed by $\tilde{W}_t$.

The Reference Portfolio

- The non-tradable target I_t is defined as the geometric fractional OU process:

$$I_t = V_0 \exp(\xi_t - \xi_0)$$

- **Property:** At the starting $t = 0$, the initial target matches the deterministic capital allocation $I_0 = V_0$.

Tradable Market

The Tradable Financial Market ($t \geq 0$)

- The market consists of $n-1$ liquid risky assets and 1 risk-free asset.
- All model coefficients $(\mu_{it}, \sigma_{it}, r_t)$ are bounded deterministic functions of time.

Risky Assets Dynamics ($i = 1, \ldots, n-1$)

$$dS_{it} = \mu_{it} S_{it} dt + \sigma_{it} S_{it} dW_{it}, \quad S_{i0} > 0$$

where the market Brownian motions W_{it} exhibit a constant correlation structure:

$$dW_{it} dW_{jt} = \rho_{ij} dt \quad (i \neq j), \quad -1 < \rho_{ij} < 1$$

Risk-Free Asset Dynamics ($i = n$)

$$dS_{nt} = r_t S_{nt} dt, \quad S_{n0} > 0$$

Information and Independence

- The vector market noise $W_t = (W_{1t}, \ldots, W_{(n-1)t})'$ is *independent* of $\tilde{W}_t$ (driving BM of the fBM).

The Portfolio Dynamics and Admissible Controls

Portfolio Process

$$dX_t = (r_t X_t + \boldsymbol{b}_t' U_t)dt + U_t' \boldsymbol{D}_{\sigma,t} d\boldsymbol{W}_t, \quad X_0 = V_0$$

where

- $U_t = (U_{1t}, \ldots, U_{(n-1)t})'$ is the control vector of capital amounts.
- $\boldsymbol{b}_t = (b_{1t}, \ldots, b_{(n-1)t})'$ is the excess return vector, where $b_{it} = \mu_{it} - r_t$.
- $\boldsymbol{D}_{\sigma,t} = \text{diag}(\sigma_{1t}, \ldots, \sigma_{(n-1)t})$ is the diagonal volatility matrix.
- $\boldsymbol{W}_t = (W_{1t}, \ldots, W_{(n-1)t})'$ is the vector of correlated Brownian motions.
- $\boldsymbol{\Sigma}_t$ is the covariance matrix with entries:

$$[\boldsymbol{\Sigma}_t]_{ij} = \begin{cases} \sigma_{it}^2, & i = j \\ \sigma_{it}\sigma_{jt}\rho_{ij}, & i \neq j \end{cases} \quad \text{where } -1 < \rho_{ij} < 1$$

Admissible Controls

- A trading strategy U_t belongs to the class $\mathcal{U}$ if it is $\mathcal{F}_t$-adapted and satisfies:

$$\int_0^T \|U_t\|^2 dt < \infty \quad \mathbb{P}\text{-a.s.}, \quad \text{for all } T > 0$$

Infinite-Horizon Pathwise Optimal Control

The Ergodic Tracking Criterion

- The objective is to minimize the (pathwise) long-run average quadratic tracking error *almost surely* ($\mathbb{P}$-a.s.) over an infinite horizon:

$$\limsup_{T\to\infty} \frac{\int_0^T \left[q(X_t - l_t)^2 + U_t' R U_t\right] dt + q_1(X_T - l_T)^2}{T} \longrightarrow \inf_{U\in\mathcal{U}}$$

where

- $q > 0$ is the continuous state tracking error penalty.
- $q_1 > 0$ is the terminal mismatch penalty weight.
- $R > 0$ is a positive definite matrix representing control costs or transaction restrictions.

Pathwise Optimization Feature

- **No Expectation Operator:** The minimization holds pathwise ($\mathbb{P}$-a.s.) for almost every realization $\omega \in \Omega$.

Market Parameter Stabilization Condition

Sufficient Condition for Market Coefficients

- We assume the existence of a parameter function $\gamma_t > 0$ such that the market coefficients satisfy the following restriction:

$$\sup_{t_0 \geq 0} \left\{ \int_{t_0}^{t} \left[2r_u + (\gamma_u^2 - 2\gamma_u) \boldsymbol{b}_u' \boldsymbol{\Sigma}_u^{-1} \boldsymbol{b}_u \right] du + \rho(t - t_0) \right\} \leq M$$

Parameters:

- $M > 0$ and $\rho > 0$ are uniform global constants; $t \geq t_0 \geq 0$.

Uniformly Bounded Case and Sharpe Metric

- This condition is satisfies if the market coefficients are uniformly bounded and the market price of risk is bounded away from zero:

$$r_t \leq \bar{r}, \quad \boldsymbol{b}_t' \boldsymbol{\Sigma}_t^{-1} \boldsymbol{b}_t \geq \underline{\lambda} > 0$$

- Choosing a constant $\gamma \in (0, 2)$ then guarantees stabilization as long as the market Sharpe metric is sufficiently large relative to the risk-free rate: $(2\gamma - \gamma^2)\underline{\lambda} > 2\bar{r}$.

Optimal Control Strategy

The Deterministic Riccati Equation

- Under the stabilization condition, there exists a unique bounded solution P_t to the differential Riccati-type equation on $[0, \infty)$:

$$\dot{P}_t + 2r_t P_t + q - P_t^2 \boldsymbol{b}_t' (\boldsymbol{R} + P_t \boldsymbol{\Sigma}_t)^{-1} \boldsymbol{b}_t = 0$$

Proposition (Uniform Positivity and MS Stability)

- The solution P_t is uniformly positive: $P_t \geq \underline{P} > 0$ for all $t \geq 0$.
- Let $A_t = r_t - P_t \boldsymbol{b}_t' (\boldsymbol{R} + P_t \boldsymbol{\Sigma}_t)^{-1} \boldsymbol{b}_t$ be the closed-loop drift. The market system retains its *uniform exponential mean-square stability* (UEMSS), satisfying:

$$\sup_{t_0 \geq 0} \left\{ \int_{t_0}^{t} \left[2A_u + P_t^2 \boldsymbol{b}_u' (\boldsymbol{R} + P_t \boldsymbol{\Sigma}_u)^{-1} \boldsymbol{\Sigma}_u (\boldsymbol{R} + P_t \boldsymbol{\Sigma}_u)^{-1} \boldsymbol{b}_u \right] du + \rho_0 (t - t_0) \right\} \leq M_0$$

for uniform constants $M_0 > 0$ and $\rho_0 > 0$.

Optimal Strategy $\boldsymbol{U}_t^*$

$$\boldsymbol{U}_t^* = -(\boldsymbol{R} + P_t \boldsymbol{\Sigma}_t)^{-1} \boldsymbol{b}_t \left(P_t X_t - Y_t \right)$$

where Y_t is the unique solution to the corresponding BSDE on $[0, \infty)$.

BSDE Representation and Malliavin Calculus

The Infinite-Horizon BSDE

- Since I_t is driven solely by $\tilde{W}_t$, the process Y_t satisfies the scalar BSDE on $\mathbb{R}^+$ without a boundary condition:

$$dY_t + [A_u Y_t + q I_t]\, dt + \tilde{Z}_t d\tilde{W}_t = 0 \tag{1}$$

Closed-Form Adapted Solution

- The unique adapted solution Y_t can be written via conditional expectations as:

$$Y_t = \mathbb{E}\left[\int_t^\infty e^{\int_t^s A_u\, du} q I_s\, ds \,\middle|\, \mathcal{F}_t^{\tilde{W}} \right] \tag{2}$$

Martingale Generator via Clark–Ocone Theorem

- Applying the standard Malliavin derivative $\mathbb{D}_t$ with respect to the underlying Brownian motion $\tilde{W}_t$, the martingale generator $\tilde{Z}_t$ is given by:

$$\tilde{Z}_t = \mathbb{E}\left[\int_t^\infty e^{\int_t^s A_u\, du} q\, \mathbb{D}_t(I_s)\, ds \,\middle|\, \mathcal{F}_t^{\tilde{W}} \right] \tag{3}$$

Closed-Form Representation of Y_t and $\tilde{Z}_t$

The Fractional OU State Variable ($s > t$):

$$\xi_s = \mu + (\xi_t - \mu)e^{-\theta(s-t)} + \sigma \int_t^s e^{-\theta(s-u)} dB_u^H$$

Closed-form for Y_t:

$$Y_t = ql_t \int_t^\infty \exp\left\{ \int_t^s A_u \, du + (\xi_t - \mu)(e^{-\theta(s-t)} - 1) + \sigma\hat{\xi}_{t,s} + \frac{1}{2}v(t,s) \right\} ds$$

Closed-form for $\tilde{Z}_t$:

$$\tilde{Z}_t = \sigma q l_t \int_t^\infty \exp\left\{ \int_t^s A_u du + (\xi_t - \mu)(e^{-\theta(s-t)} - 1) + \sigma\hat{\xi}_{t,s} + \frac{1}{2}v(t,s) \right\} \mathcal{K}_H(t,s,t)\, ds$$

where the memory terms and kernels are defined as:

- $\hat{\xi}_{t,s} = \int_{-\infty}^t \mathcal{K}_H(t,s,\tau) d\tilde{W}_\tau$ is the accumulated non-Markovian memory.
- $v(t,s) = \sigma^2 \int_t^s \left(\mathcal{K}_H(t,s,\tau)\right)^2 d\tau$ is the deterministic conditional variance.
- $\mathcal{K}_H(t,s,\tau) = \int_t^s e^{-\theta(s-u)} \frac{\partial K_H(u,\tau)}{\partial u} du$ is the prediction memory kernel.

The Optimal Portfolio Process Structure

Optimal Portfolio

- The closed-loop SDE for the optimal portfolio X_t^*:

$$dX_t^* = \left[A_t X_t^* + \boldsymbol{b}_t'(\boldsymbol{R} + P_t\boldsymbol{\Sigma}_t)^{-1}\boldsymbol{b}_t Y_t\right] dt - (P_t X_t^* - Y_t)\,\boldsymbol{b}_t'(\boldsymbol{R}+P_t\boldsymbol{\Sigma}_t)^{-1}\boldsymbol{D}_{\sigma,t}\,d\boldsymbol{W}_t$$

where $A_t = r_t - P_t\boldsymbol{b}_t'(\boldsymbol{R} + P_t\boldsymbol{\Sigma}_t)^{-1}\boldsymbol{b}_t$ is the stable closed-loop drift coefficient.

- **Inhomogeneous Linear SDE:** The optimal portfolio is governed by a linear forward SDE with respect to X_t^*, containing a non-Markovian random input driven by the process Y_t.
- **Stochastic Volatility Structure:** The portfolio's effective diffusion vector is modulated by the internal feedback mechanism of the system, parameterized directly by the distance between the scaled state and the fractional memory target: $(P_t X_t^* - Y_t)$.

Proposition (Subexponential Growth)

- Under the established UEMSS parameter condition, there exists a constant $0 < \delta < 1$ such that the optimal portfolio process satisfies:

$$\lim_{T\to\infty} \frac{(X_T^*)^2}{T^\delta} = 0 \quad \mathbb{P}\text{-almost surely (a.s.)}$$

Pathwise Ergodic Optimality

Theorem (Pathwise LQ Tracking Solution)

Under the established UEMSS condition, the control strategy $\boldsymbol{U}_t^* = -(\boldsymbol{R} + P_t\boldsymbol{\Sigma}_t)^{-1}\boldsymbol{b}_t\,(P_t X_t^* - Y_t)$ *is the pathwise ergodic optimal control.*

The Minimal Total Cost J^*

- The long-run pathwise average tracking error limit converges strictly to a deterministic constant $\mathbb{P}$-almost surely (a.s.), where the minimal cost J^* reads:

$$J^* = \limsup_{T\to\infty} \frac{1}{T}\int_0^T \left[q\,\mathbb{E}[l_t^2] - \mathbb{E}[Y_t^2]\,\boldsymbol{b}_t'(\boldsymbol{R} + P_t\boldsymbol{\Sigma}_t)^{-1}\boldsymbol{b}_t\right] dt \quad \mathbb{P}\text{-a.s.} \tag{4}$$

Idea of the Proof

Fix a competing control $\boldsymbol{U} \in \mathcal{U}$ and estimate:

$$\begin{aligned} J_T(\boldsymbol{U}^*) - J_T(\boldsymbol{U}) \le c_1\Big(X_T^{*2} + Y_T^2 + l_T^2 - \int_0^T \Big[q(X_t - X_t^*)^2 + (\boldsymbol{U}_t - \boldsymbol{U}_t^*)'\boldsymbol{R}(\boldsymbol{U}_t - \boldsymbol{U}_t^*)\Big] dt \\ + \mathcal{M}_{1T}(X_t - X_t^*, X_t^*) + \mathcal{M}_{2T}(\boldsymbol{U}_t - \boldsymbol{U}_t^*, X_t^*) \\ + \mathcal{M}_{3T}(X_t - X_t^*, \tilde{Z}_t)\Big). \end{aligned}$$

Apply the LIL for martingales $\mathcal{M}_{iT}$ and sublinear growth rates of the terminal cost, and pass to the ergodic limit.

Analytical Reduction of the Minimal Tracking Cost

Simplified Pathwise Ergodic Criterion

- Under stationary transformation:

$$J^* = q \cdot C_l \cdot (V_0 e^{-\xi_0})^2 \cdot \limsup_{T\to\infty} \frac{1}{T} \int_0^T \left[1 - q \cdot \hat{\mathcal{K}}_H(t)\, \boldsymbol{b}_t' (\boldsymbol{R} + P_t \boldsymbol{\Sigma}_t)^{-1} \boldsymbol{b}_t\right] dt \quad \mathbb{P}\text{-a.s.}$$

where

- C_l is the log-normal scaling constant:

$$C_l = \exp\left(2\mu + 2\nu_H^2\right), \quad \nu_H^2 = \frac{\sigma^2 \Gamma(2H+1)\sin(\pi H)}{2\pi\theta^{2H}} \int_{-\infty}^{+\infty} \frac{|x|^{1-2H}}{1+x^2}\, dx$$

- $\hat{\mathcal{K}}_H(t)$ is the decoupled deterministic memory scaling kernel:

$$\hat{\mathcal{K}}_H(t) = \int_t^\infty \int_t^\infty e^{\int_t^s A_u du + \int_t^\tau A_u du} \cdot \exp\left(\mathrm{Cov}(\hat{\xi}_{s|t}, \hat{\xi}_{\tau|t}) - \nu_H^2\right) ds\, d\tau$$

Long-Run Asymptotic Analysis of the Information Gain

The Ergodic Interaction Structure

$$\Delta J^* = \liminf_{T\to\infty} \frac{1}{T} \int_0^T q \cdot \hat{\mathcal{K}}_H(t)\, \boldsymbol{b}_t'(\boldsymbol{R} + P_t \boldsymbol{\Sigma}_t)^{-1} \boldsymbol{b}_t \, dt \quad \mathbb{P}\text{-a.s.}$$

Uniform Bounds for the Memory Kernel (α_1, α_2 are the maximal and minimal stability rates)

$$\frac{e^{-2\nu_H^2}}{\alpha_1^2} \leq \hat{\mathcal{K}}_H(t) \leq \frac{1}{\alpha_2^2}$$

Corollary (UEMSS Structural Feedback Alignment)

- The condition for a strict tracking improvement ($\Delta J^* > 0$) given by:

$$\liminf_{T\to\infty} \frac{1}{T} \int_0^T \boldsymbol{b}_t' \boldsymbol{\Sigma}_t^{-1} \boldsymbol{b}_t \, dt > 0$$

holds as a direct consequence of the closed-loop UEMSS system property under the optimal portfolio control.

Thank you for your attention!

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